Mathletes Formulas

Sheldon Mathletes

2026-2027

Table of Contents

1. Polynomials 2
1.1. Linear Equations 2
1.2. Quadratics 2
1.3. General Polynomials 2
1.4. Rational Functions 3

2. Trigonometry 4
2.1. Functions 4
2.2. Identities 5
2.3. Laws 6

3. Graphs and Geometry 7
3.1. Lines and Points 7
3.2. Shapes 7
3.3. Graph Manipulation 8

4. Miscellaneous 8
4.1. Inverse Functions 8
4.2. Series 8
4.3. Complex Numbers 9
4.4. Exponential and Logarithmic Properties 9

1. Polynomials

1.1. Linear Equations

Slope-Intercept $y = mx + b$ **Slope** $m = \dfrac{\Delta y}{\Delta x} = \dfrac{y_2 - y_1}{x_2 - x_1}$

Point-Slope $y - y_1 = m(x - x_1)$ **Point-Point** $y - y_1 = \dfrac{y_2 - y_1}{x_2 - x_1}(x - x_1)$

Parallel lines have equivalent slopes, while perpendicular lines have slopes that are negative reciprocals of one another ($m \perp -\frac{1}{m}$)

1.2. Quadratics

Quadratic Equations

Standard Form $ax^2 + bx + c$

Vertex Form $a(x - h)^2 + k$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Let $D = b^2 - 4ac$,

1. $D > 0 \implies$ 2 real solutions
2. $D = 0 \implies$ 1 real solution
3. $D < 0 \implies$ no real solutions

Minimum and Maximum Values

Consider the function $f(x) = ax^2 + bx + c$ with vertex $\left(-\frac{b}{2}a, f\left(-\frac{b}{2}a\right)\right)$

1. $a > 0 \implies f$ has a *minimum* at $x = -\frac{b}{2}a$. The minimum value is $f\left(-\frac{b}{2}a\right)$.
2. $a < 0 \implies f$ has a *maximum* at $x = -\frac{b}{2}a$. The maximum value is $f\left(-\frac{b}{2}a\right)$.

1.3. General Polynomials

Let n be a non-negative integer and let $a_n, a_{n-1}, ..., a_2, a_1, a_0$ be real numbers such that $a_n \neq 0$. The function:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_2 x^2 + a_1 x + a_0$$

is a polynomial function of x with degree n.

Linear Factorization Theorem

If $f(x)$ is a polynomial of degree n, where $n > 0$, then $f(x)$ has precisely n linear factors

$$f(x) = a_n(x - c_1)(x - c_2)...(x - c_n)$$

where $c_1, c_2, ..., c_n$ are complex numbers.

Rational Zero Test

If the polynomial has *integer* coefficients, then every rational zero of f has the form $\frac{p}{q}$, where p and q have no commons factors other than 1, and

$p =$ a factor of the constant term a_0 $q =$ a factor of the leading coefficient a_n

Conjugate Pairs

Let f be a polynomial function with *real coefficients*. If $a + bi$, where $b \neq 0$, is zero of f, the complex conjugate $a - bi$ is also a zero of f.

For polynomials with *integer coefficients*, $a + \sqrt{b}$ being a zero implies $a - \sqrt{b}$ is also a zero.

Roots

Let f be the polynomial

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + ... + a_2 x^2 + a_1 x + a_0$$

The sum of all roots $= -\dfrac{a_n}{a_{n-1}}$ The product of all roots $= (-1)^n \dfrac{a_0}{a_n}$

1.4. Rational Functions

Let f be the rational function

$$f(x) = \frac{N(x)}{D(x)} = \frac{a_n x^n + a_{n-1} x^{n-1} + ... + a_1 x + a_0}{b_m x^m + b_{m-1} x^{m-1} + ... + b_1 x + b_0}$$

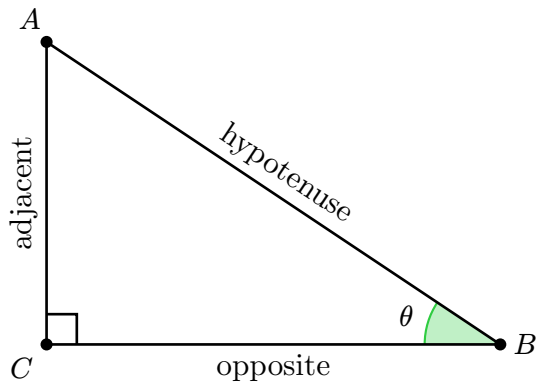
where $N(x)$ and $D(x)$ have no common factors.

1. The graph of f has *vertical* asymptotes at the zeros of $D(x)$.
2. The graph of f has at most one *horizontal asymptote* determined by comparing the degrees of $N(x)$ and $D(x)$.
    a. $n < m \implies$ the line $y = 0$ is the horizontal asymptote.
    b. $n = m \implies$ the line $y = \frac{a_n}{b_m}$ is the horizontal asymptote.
    c. $n > m \implies$ there is no horizontal asymptote.

Suppose $N(x)$ and $D(x)$ had some common factor $(x - c)$. The graph of f would then have a hole at $(c, f(c))$

2. Trigonometry

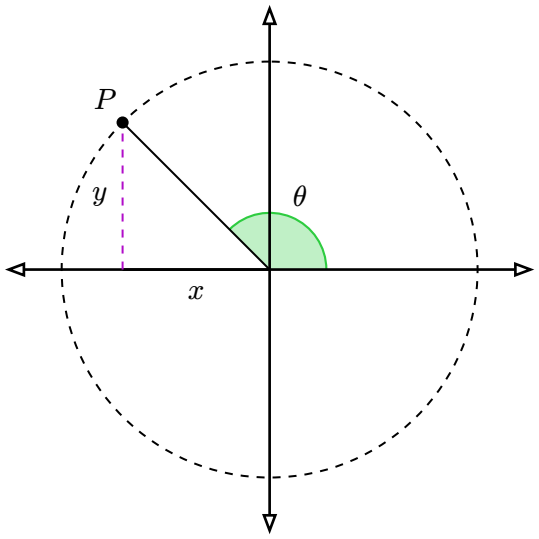
2.1. Functions



$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} \qquad \csc\theta = \frac{\text{hypotenuse}}{\text{opposite}}$$

$$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} \qquad \sec\theta = \frac{\text{hypotenuse}}{\text{adjacent}}$$

$$\tan\theta = \frac{\text{opposite}}{\text{adjacent}} \qquad \cot\theta = \frac{\text{adjacent}}{\text{opposite}}$$



$$\sin\theta = \frac{y}{1} = y \qquad \csc\theta = \frac{1}{y}$$

$$\cos\theta = \frac{x}{1} = x \qquad \sec\theta = \frac{1}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{x}{y}$$

Function	Domain	Range	Period
sin	$\mathbb{R}$	$[-1, 1]$	2π
cos	$\mathbb{R}$	$[-1, 1]$	2π
tan	$\left\{x \mid x \in \mathbb{R}, x \neq \frac{\pi}{2} + n\pi\right\}$ *	$\mathbb{R}$	π
csc	$\{x \mid x \in \mathbb{R}, x \neq n\pi\}$ *	$(-\infty, -1] \cup [1, \infty)$	2π
sec	$\left\{x \mid x \in \mathbb{R}, \pi \neq \frac{\pi}{2} + n\pi\right\}$ *	$(-\infty, -1] \cup [1, \infty)$	2π
cot	$\{x \mid x \in \mathbb{R}, x \neq n\pi\}$ *	$\mathbb{R}$	π

Table 1: Functions (* $\forall n \in \mathbb{Z}$)

Function	Domain	Range
$\sin^{-1}$ / arcsin	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$\cos^{-1}$ / arccos	$[-1, 1]$	$[0, \pi]$
$\tan^{-1}$ / arctan	$\mathbb{R}$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$\csc^{-1}$ / arccsc	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$
$\sec^{-1}$ / arcsec	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
$\cot^{-1}$ / arccot	$\mathbb{R}$	$(0, \pi)$

Table 2: Inverse Functions

	0°	30°	45°	60°	90°	180°	270°	360°
sin	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	0	-1	0
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0	-1	0	1
cos	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	**undef**	0	**undef**	0

Table 3: Common Values

2.2. Identities

Radians and Degrees

$$n^\circ = \frac{\pi}{180} n \text{ rad}$$
$$m \text{ rad} = \left(\frac{180}{\pi} m\right)^\circ$$

Reciprocals

$$\csc\theta = (\sin\theta)^{-1} \quad \sin\theta = (\csc\theta)^{-1}$$
$$\sec\theta = (\cos\theta)^{-1} \quad \cos\theta = (\sec\theta)^{-1}$$
$$\cot\theta = (\tan\theta)^{-1} \quad \tan\theta = (\cot\theta)^{-1}$$

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$
$$\tan^2\theta + 1 = \sec^2\theta$$
$$1 + \cot^2\theta = \csc^2\theta$$

Even/Odd

$$\sin(-\theta) = -\sin\theta \quad \csc(-\theta) = -\csc\theta$$
$$\cos(-\theta) = \cos\theta \quad \sec(-\theta) = \sec\theta$$
$$\tan(-\theta) = -\tan\theta \quad \cot(-\theta) = -\cot\theta$$

Cofunction

$$\sin(90^\circ - \theta) = \cos\theta \quad \cos(90^\circ - \theta) = \sin\theta$$
$$\sec(90^\circ - \theta) = \csc\theta \quad \csc(90^\circ - \theta) = \sec\theta$$
$$\tan(90^\circ - \theta) = \cot\theta \quad \cot(90^\circ - \theta) = \tan\theta$$

Sum and Difference

$$\sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\tan\beta}$$

Double Angle

$$\begin{aligned}\sin 2\theta &= 2\sin\theta\cos\theta \\ \cos 2\theta &= \cos^2\theta - \sin^2 x \\ &= 2\cos^2\theta - 1 \\ &= 1 - 2\sin^2\theta \\ \tan 2\theta &= \frac{2\tan\theta}{1-\tan^2\theta}\end{aligned}$$

Sum to Product

$$\begin{aligned}\sin\alpha + \sin\beta &= 2\sin\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \\ \sin\alpha - \sin\beta &= 2\cos\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right) \\ \cos\alpha + \cos\beta &= 2\cos\left(\frac{\alpha+\beta}{2}\right)\cos\left(\frac{\alpha-\beta}{2}\right) \\ \cos\alpha - \cos\beta &= -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)\end{aligned}$$

Half Angle

$$\sin\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1-\cos\theta}{2}}$$
$$\cos\left(\frac{\theta}{2}\right) = \pm\sqrt{\frac{1+\cos\theta}{2}}$$
$$\tan\left(\frac{\theta}{2}\right) = \frac{\sin\theta}{1+\cos\theta} = \frac{1-\cos\theta}{\sin\theta}$$

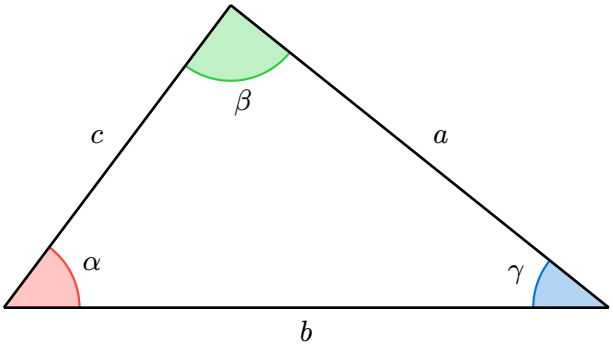
Product to Sum

$$\sin\alpha\sin\beta = \frac{\cos(\alpha-\beta) - \cos(\alpha+\beta)}{2}$$
$$\cos\alpha\cos\beta = \frac{\cos(\alpha+\beta) + \cos(\alpha-\beta)}{2}$$
$$\sin\alpha\cos\beta = \frac{\sin(\alpha+\beta) + \sin(\alpha-\beta)}{2}$$
$$\cos\alpha\sin\beta = \frac{\sin(\alpha+\beta) - \sin(\alpha-\beta)}{2}$$

Power Reduction

$$\sin^2\theta = \frac{1-\cos 2\theta}{2}$$
$$\cos^2\theta = \frac{1+\cos 2\theta}{2}$$
$$\tan^2\theta = \frac{1-\cos 2\theta}{1+\cos 2\theta}$$

2.3. Laws



$$\text{Area} = \frac{1}{2}bc\sin\alpha = \frac{1}{2}ac\sin\beta = \frac{1}{2}ab\sin\gamma$$

Law of Sines

$$\frac{a}{\sin\alpha} = \frac{b}{\sin\beta} = \frac{c}{\sin\gamma}$$
$$\frac{\sin\alpha}{a} = \frac{\sin\beta}{b} = \frac{\sin\gamma}{c}$$

Mollweide's Formula

$$\frac{a+b}{c} = \frac{\cos\left(\frac{1}{2}(\alpha-\beta)\right)}{\sin\left(\frac{1}{2}\gamma\right)}$$

Law of Cosines

$$a^2 = b^2 + c^2 - 2bc\cos\alpha$$
$$b^2 = a^2 + c^2 - 2ac\cos\beta$$
$$c^2 = a^2 + b^2 - 2ab\cos\gamma$$
$$\cos\alpha = \frac{b^2 + c^2 - a^2}{2bc}$$
$$\cos\beta = \frac{a^2 + c^2 - b^2}{2bc}$$
$$\cos\gamma = \frac{a^2 + b^2 - c^2}{2bc}$$

Law of Tangents

$$\frac{a-b}{a+b} = \frac{\tan\left(\frac{1}{2}(\alpha-\beta)\right)}{\tan\left(\frac{1}{2}(\alpha+\beta)\right)}$$
$$\frac{b-c}{b+c} = \frac{\tan\left(\frac{1}{2}(\beta-\gamma)\right)}{\tan\left(\frac{1}{2}(\beta+\gamma)\right)}$$
$$\frac{a-c}{a+c} = \frac{\tan\left(\frac{1}{2}(\alpha-\gamma)\right)}{\tan\left(\frac{1}{2}(\alpha+\gamma)\right)}$$

3. Graphs and Geometry

3.1. Lines and Points

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Pythagorean Theorem

$$a^2 + b^2 = c^2 \implies c = \sqrt{a^2 + b^2}$$

Perpendicular Bisector

$$2(x_2 - x_1)x + 2(y_2 - y_1)y = x_2^2 + y_2^2 - x_1^2 - y_1^2$$

3.2. Shapes

Circles

$(x-m)^2 + (y-n)^2 = r^2$ Area $= \pi r^2$ Circumference $= 2\pi r$

Regular Polygons

Sum of Interior Angles $= 180(n-2)°$ Area $= \frac{1}{2} \cdot \text{Apothem} \cdot \text{Perimeter}$

Volumes

$V_{\text{Cylinder}} = \pi r^2 h$ $V_{\text{Cone}} = \frac{1}{3}\pi r^2 h$ $V_{\text{Sphere}} = \frac{4}{3}\pi r^3$ $V_{\text{Pyramid}} = \frac{1}{3}Bh$

Triangle Similarity

Two triangles $\triangle ABC$ and $\triangle DEF$ are similar if

SSS

$$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$$

SAS

$$\frac{AB}{DE} = \frac{AC}{DF} \text{ and } m\angle A = m\angle D$$

AA

$$m\angle A = m\angle D$$
$$m\angle B = m\angle E$$

3.3. Graph Manipulation

Transformation	Conditions	Description
$g(x) = f(x \pm c)$	$(+)$ $(-)$	shift left c units shift right c units
$g(x) = f(x) \pm c$	$(+)$ $(-)$	shift left c units shift right c units
$g(x) = cf(x)$	$c < 1$ $c > 1$	vertical compress by a factor of c vertical stretch by a factor of c
$g(x) = f(cx)$	$c < 1$ $c > 1$	horizontal stretch by a factor of c horizontal compress by a factor of c
$g(x) = -f(x)$	**N/A**	reflect about the x-axis
$g(x) = f(-x)$	**N/A**	reflect about the y-axis

4. Miscellaneous

4.1. Inverse Functions

Let f and g be two function such that

$f(g(x)) = x$ for all x in the domain of g $\qquad$ $g(f(x)) = x$ for all x in the domain of f

Under these conditions, the function g is the *inverse function* of the function f (and vice-versa). The function g can then be denoted as f^{-1}. So,

$$f(f^{-1}(x)) = x \qquad f^{-1}(f(x)) = x$$

The domain of f must be equal to the range of f^{-1}, and the range of f must be equal to the domain of f^{-1}.

4.2. Series

Arithmetic

$$S_n = \frac{n}{2}(a_1 + a_n)$$

$$a_n = a_1 + (n-1)d$$

$$S_n = \frac{n}{2}(2a_1 + (n-1)d)$$

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$$

Geometric

$$S_n = \frac{a_1(1-r^n)}{1-r}, r \neq 0$$

$$\sum_{i=1}^{\infty} a_i = \frac{a_1}{1-r}, |r| < 1 *$$

$$a_n = a_1 r^{n-1}$$

* where there are infinitely many a terms

4.3. Complex Numbers

Let a and b be real numbers. The number $a + bi$ is a *complex number* written in *standard form*. The number a is the *real part* and the number bi is the *imaginary part* of the complex number.

When $b = 0$, the number $a + bi$ is a real number. When $b \neq 0$, the number $a + bi$ is an *imaginary number*. A number of the form bi, where $b \neq 0$, is a *pure imaginary number*.

$$i^1 = i, i^2 = -1, i^3 = -i, i^4 = 1, i^n = i^{n \bmod 4}$$

4.4. Exponential and Logarithmic Properties

Exponential

$$a^0 = 1$$
$$a^m \cdot a^n = a^{m+n}$$
$$\frac{a^m}{a^n} = a^{m-n}$$
$$(ab)^n = a^n \cdot b^n$$
$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$
$$(a^m)^n = a^{m \cdot n}$$
$$a^{\frac{1}{n}} = \sqrt[n]{a}$$
$$a^m = a^n \implies m = n$$

Logarithmic

$$\log_n 1 = 0$$
$$\log_n n = 1$$
$$\log x = \log_{10} x$$
$$\ln x = \log_e x$$
$$\log_n a = b \implies a = n^b$$
$$n^{\log_n x} = x$$
$$\log_n n^x = x$$
$$\log_n a^b = b \log_n a$$
$$\log_{n^b} a = \frac{1}{b} \log_n a$$
$$\log_n ab = \log_n a + \log_n b$$
$$\log_n \frac{a}{b} = \log_n a - \log_n b$$
$$\log_n a = \log_n b \implies a = b$$